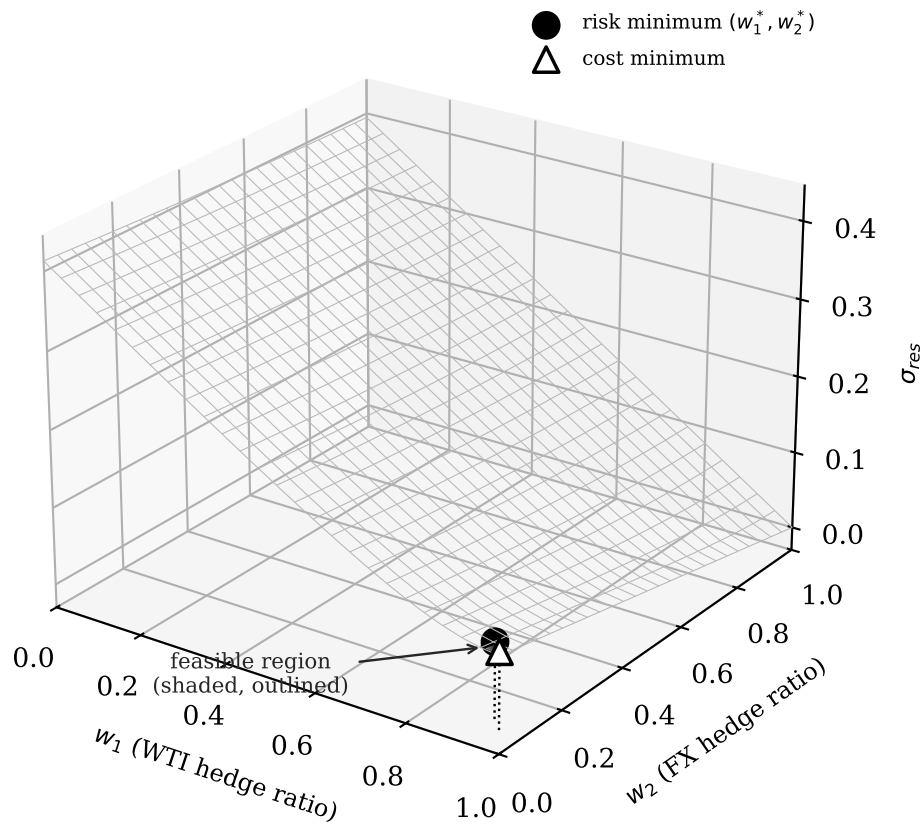


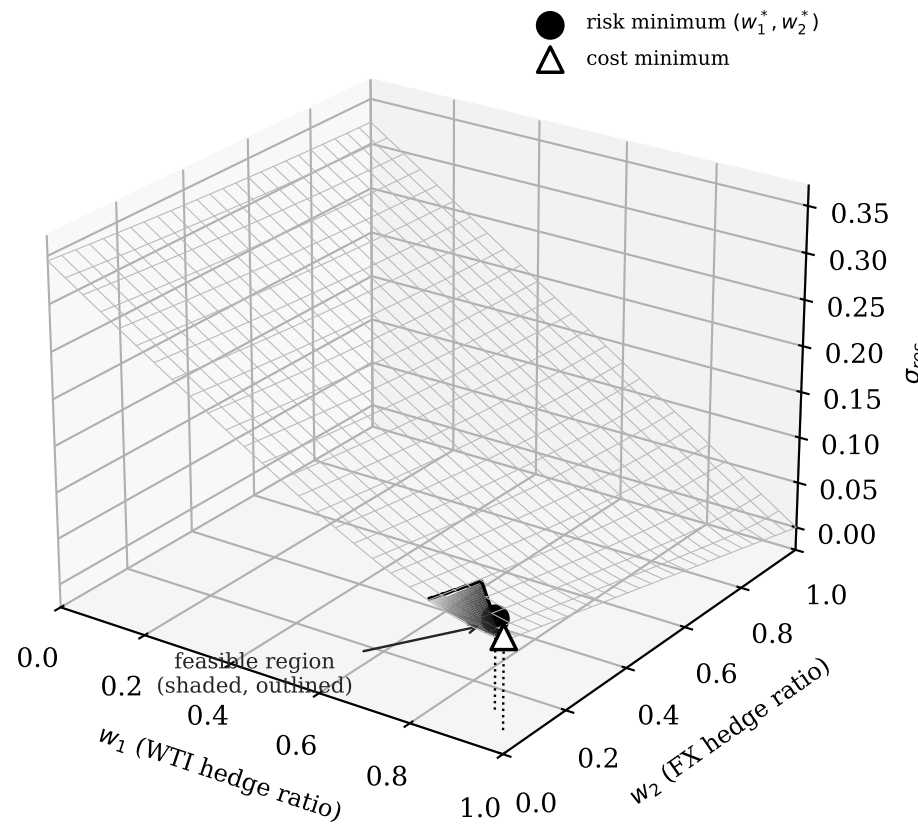
Residual-volatility surface $\sigma_{res}(w_1, w_2)$: feasible region (all four constraints) shaded, dark = low risk; infeasible region wireframe only. Top: full unit square. Bottom: 3-D zoom.

European (Black-76 / GK premiums)



zoom onto the feasible region

American (LSMC Shapley premiums)



zoom onto the feasible region

